

Generative models as distribution matching

Original $\xrightarrow{\text{Forward}}$ Degraded $\xrightarrow{\text{Reverse}}$ Generated

Noise



Input



DSBM

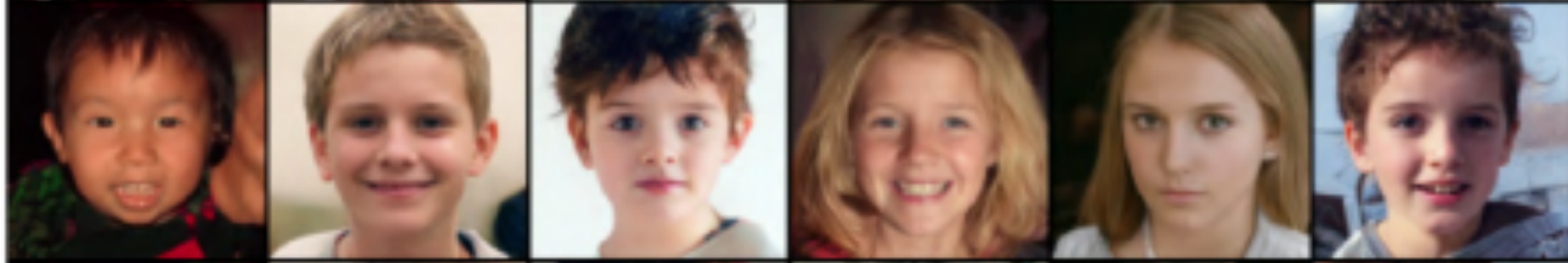
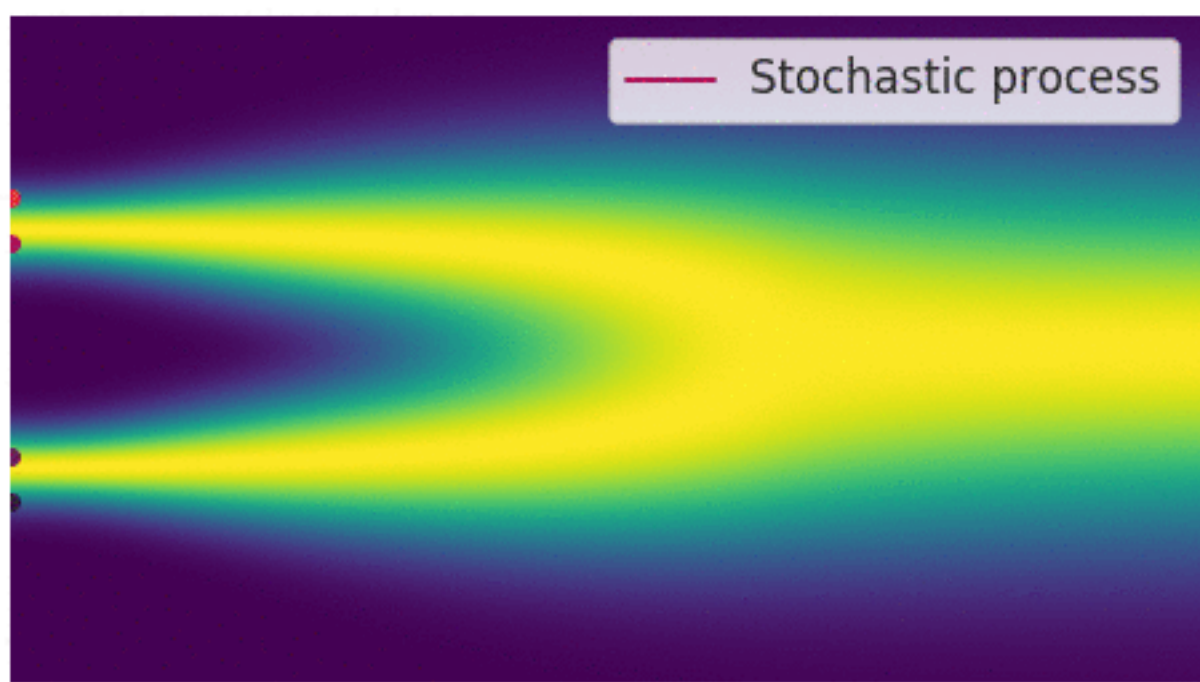


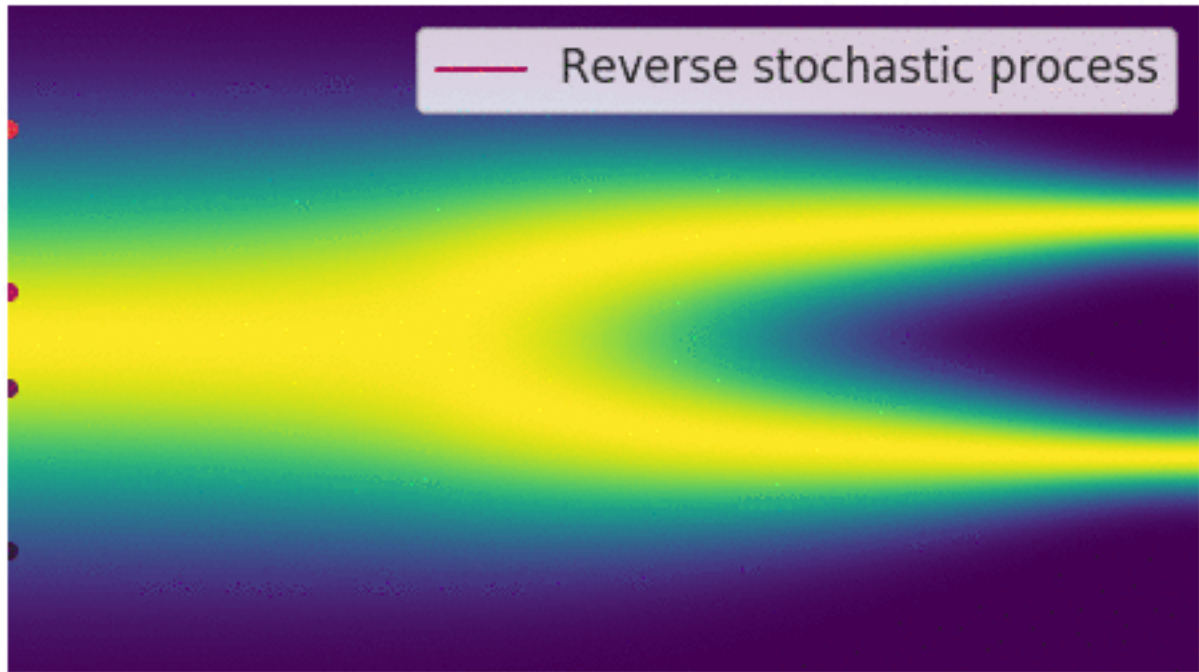
Image generation:

image  noise

Image matching:

image1 $\longleftrightarrow$ image2





Forward: image $\rightarrow$ noise

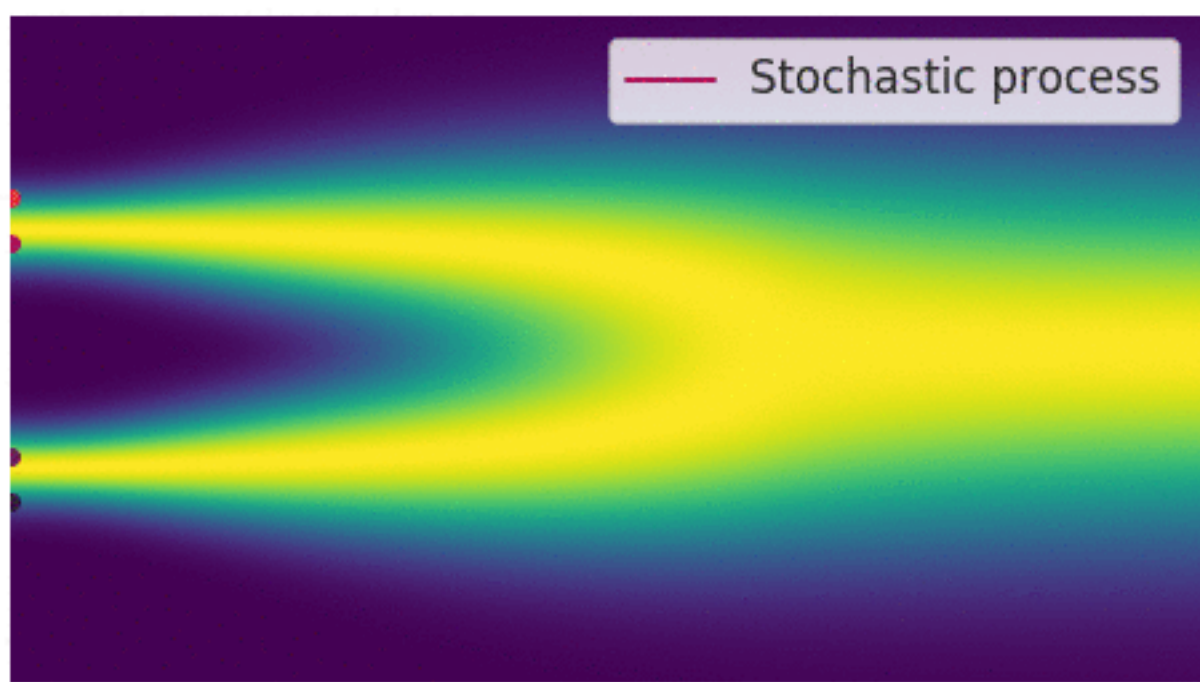
backward: noise $\rightarrow$ image

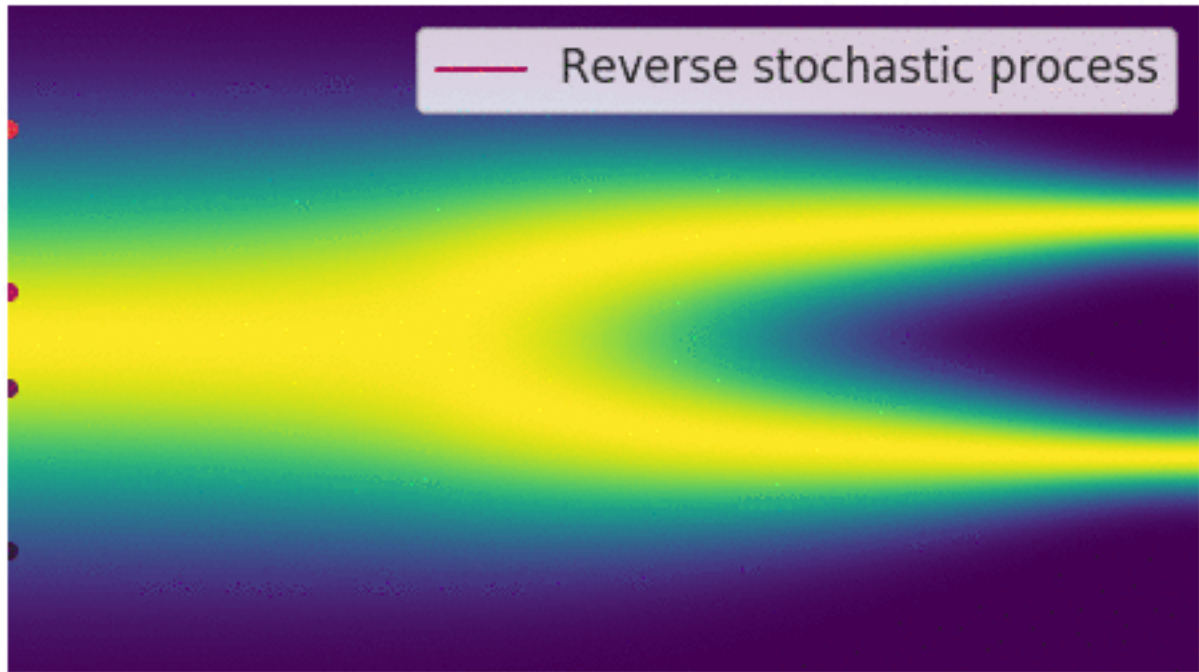
Others:

- VLM: image $\longleftrightarrow$ text

- point clouds 1 $\longleftrightarrow$ clouds 2

[song et al. 2021]





Generative models as distribution matching

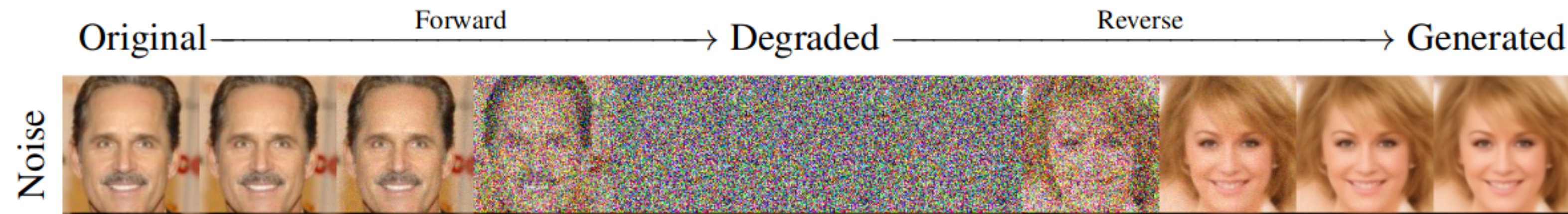


Image generation:
image $\longleftrightarrow$ noise

Forward: image $\rightarrow$ noise

backward: noise $\rightarrow$ image

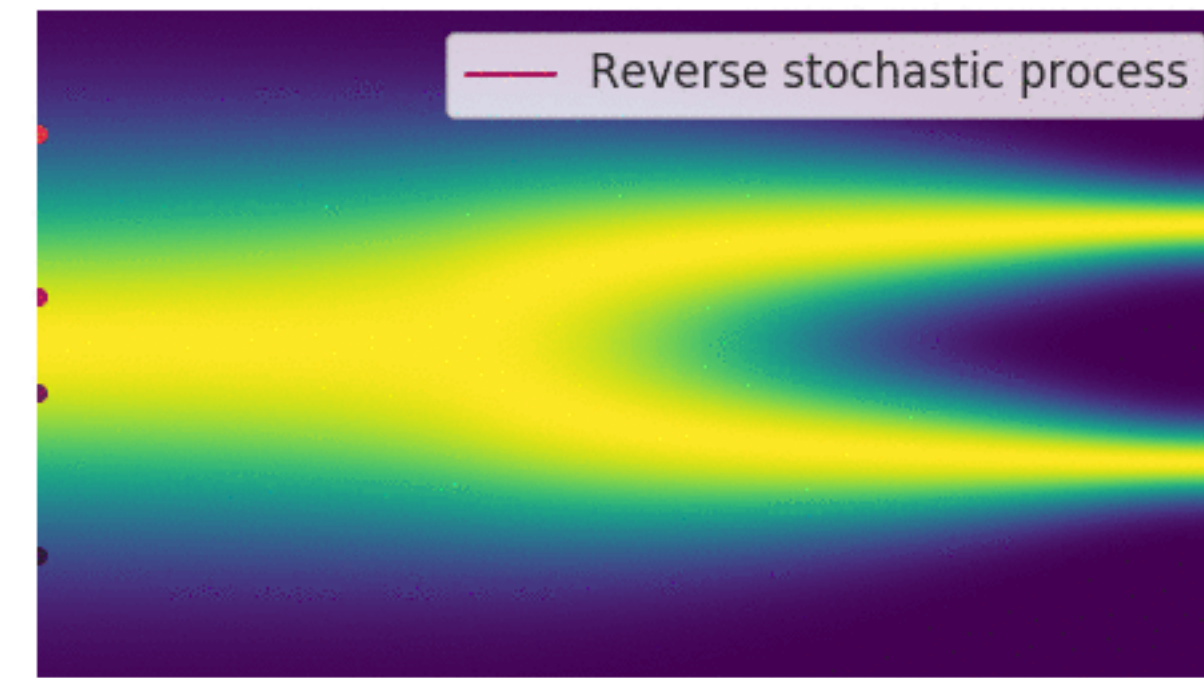
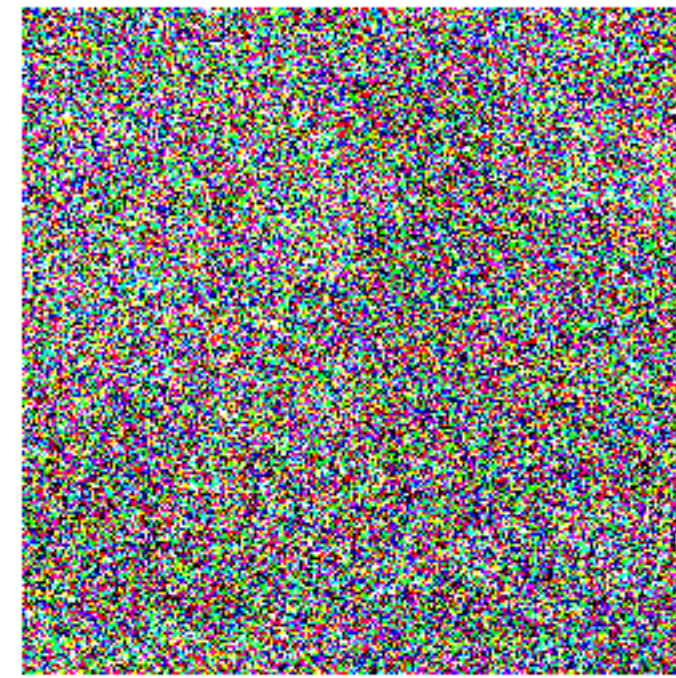
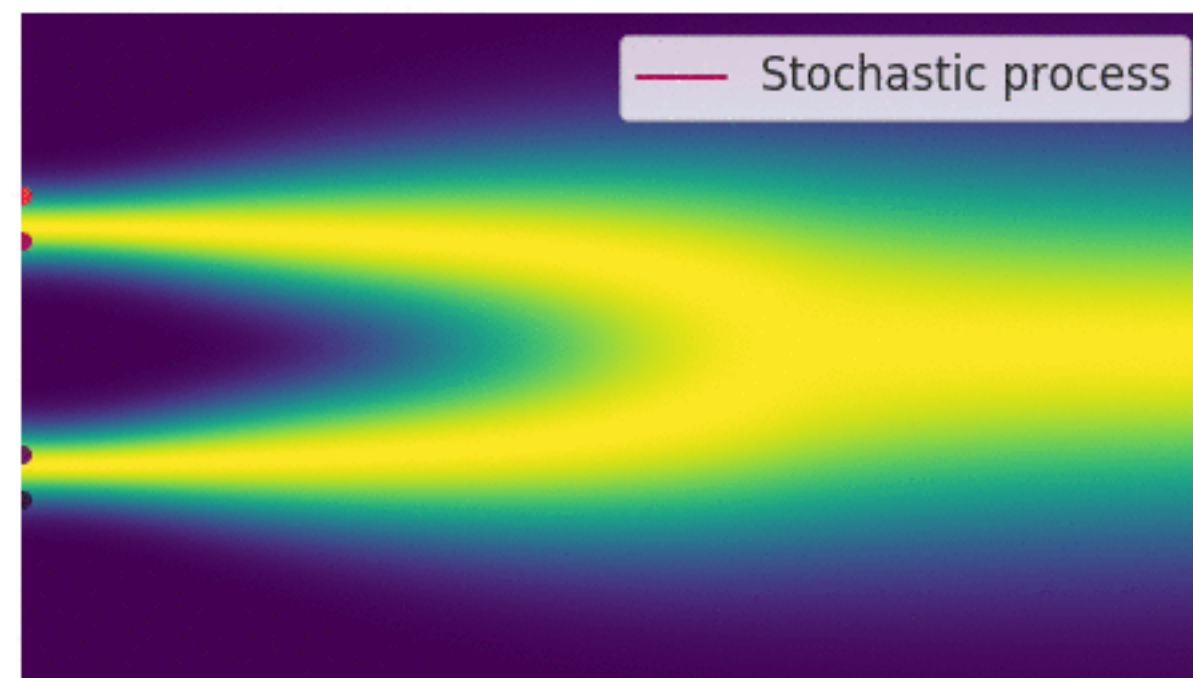


Image matching:
image1 $\longleftrightarrow$ image2



Others:

- VLM: image $\longleftrightarrow$ text
- point clouds 1 $\longleftrightarrow$ clouds 2

[Song et al. 2021]

Schrödinger's bridge (or dynamic optimal transport) unifies modern generative models

- Of the many possible (unlikely) ways, which one is the most likely? [Lénoard 2014]

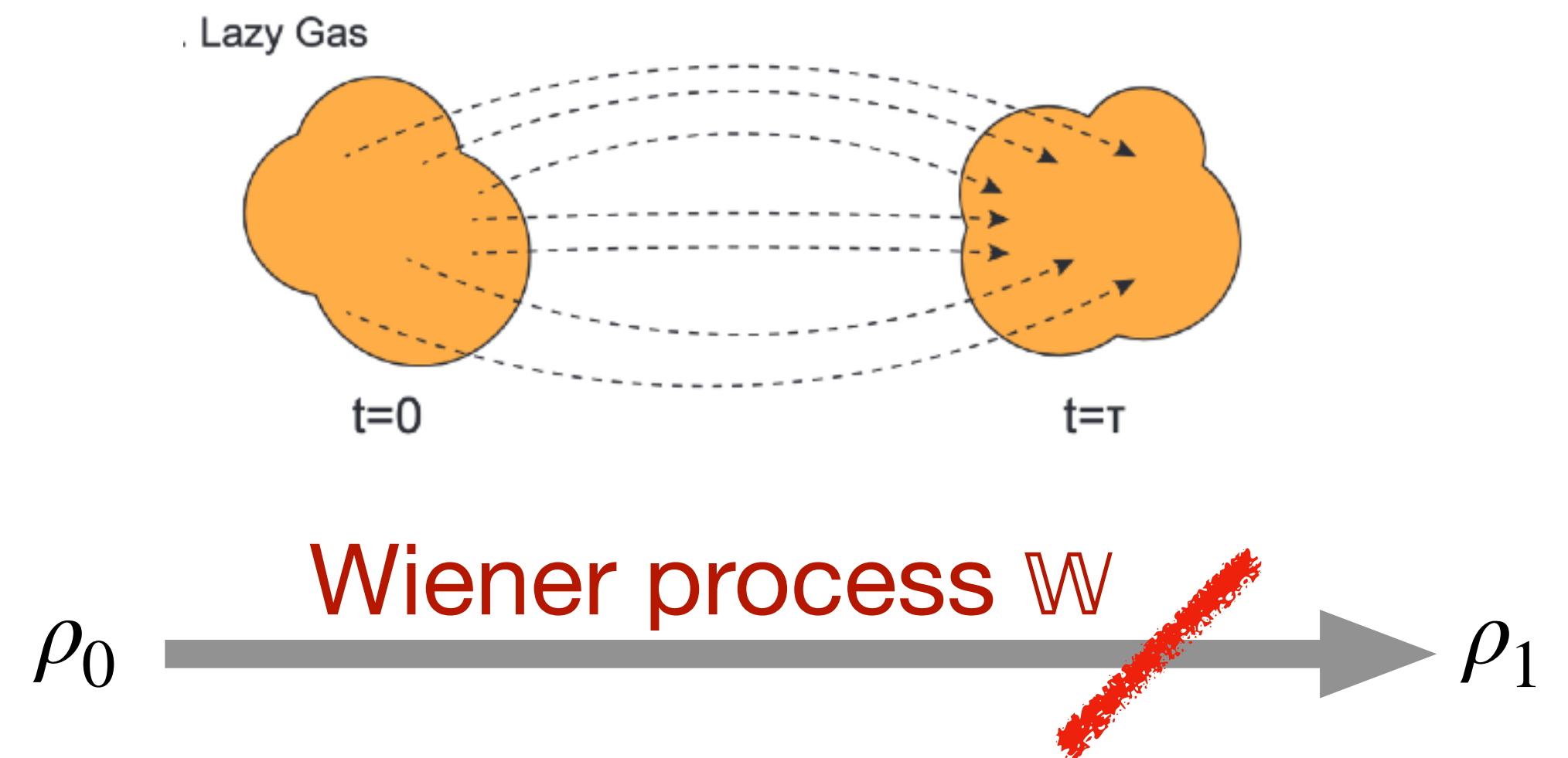
$$\min D_{KL}(\mathbb{P} \parallel \mathbb{W}) \text{ s.t. } \mathbb{P}_0 = \rho_0, \mathbb{P}_1 = \rho_1$$

- A dynamics formulation of entropic-regularized **optimal transport** [Vallani 2009]

$$\min_{\pi \in \Pi(\rho_0, \rho_1)} \int_{\mathbb{R}^n \times \mathbb{R}^n} \frac{1}{2} \|x_0 - x_1\|^2 d\pi(x_0, x_1) + \sigma^2 D_{KL}(\pi \parallel \rho_0 \otimes \rho_1)$$

- A Stochastic optimal control (SOC) formulation

$$\min_{b_t} \mathbb{E} \left[\frac{1}{2} \int_0^1 \|b(t, X_t)\|^2 dt \right], \quad \text{s.t.} \quad \begin{cases} dX_t = [g_t b(t, X_t)] dt + g_t dW_t, \\ X_0 \sim \rho_0, X_1 \sim \rho_1. \end{cases}$$



Variational approach

SDE approach

ODE approach